Decode-Position Channel Drift in Long Reasoning Traces

Anonymous authors
Paper under double-blind review

Abstract

Quamba2 argues that SSM activations exhibit channel-order preservation and activation persistence, properties that support offline per-channel quantization for Mamba-family models. We test the corresponding assumption at reasoning-scale decode horizons by measuring decode-position channel-set drift: whether channels in the top 1% by magnitude early in decode remain in the same protected set later. Granite-4.0 hybrids show large drift under frozen gates: the original rank-drift metric is 0.8178385416666667 on Granite-4.0-H-Tiny and 0.843165650406504 on Granite-4.0-H-Small. The phenomenon also appears outside the Granite family: Nemotron-3-Nano gives 0.820809591642925, DeepSeek-R1-Distill-Qwen-1.5B gives 0.8393787202380952, and Falcon-H1 gives 0.7833543771043772. A stricter set-membership decomposition shows that the systems-relevant component is not only within-set rank shuffling: strict set-leaving is 0.566234756098 on Granite-Small, 0.533713200380 on Nemotron-3, 0.670572916667 on DeepSeek, and 0.673611111111 on Falcon-H1. We then test static, position-conditioned, scale-table, smoothed, and cross-tensor W4A16 protection interventions. All fail their preregistered gates, but with different failure modes: M2 and M10 are beaten by random-bin controls, M11 is not beaten by its random-walk control yet recovers only 0.048299284138 median of the BF16-vs-static gap, and M18 activation+K coupling beats K-only and random-coupled controls but still has negative median recovery (-0.343590844126). The DecDEC algorithmic reactive baseline is less damaging than a random reactive control but also has negative median recovery (-0.070034781258). The resulting paper is a measurement and mechanism paper, not a positive quantization method: boundary discontinuities are actively harmful, smoothing removes that harm but does not recover quality, and cross-tensor coupling has diagnostic signal without reliable recovery. The remaining mechanism question is whether future methods need larger protected budgets, stable-core targeting, or a way to address long-horizon compound error.

1 Introduction

Quamba2 argues that SSM activations have channel-order preservation and activation persistence, properties that motivate offline per-channel quantization and channel clustering for Mamba1/Mamba2 backbones (Chiang et al., 2025). We show that the corresponding protected-set assumption breaks down at reasoning-scale decode horizons in our W4A16 setting: across three hybrid reasoning LLMs (Granite-4.0-H-Small, Nemotron-3-Nano, Falcon-H1) and one pure-Transformer reference (DeepSeek-R1-Distill-Qwen-1.5B), 53–67% of top-1% magnitude channels at calibration positions strictly leave the top-1% set by the final measured decode position. This is not merely within-set rank shuffling; it is strict set membership change. The distinction matters for deployment. Static protected-channel maps, rotations, per-channel formats, and calibration-time outlier policies are easiest to engineer when the protected set remains stable. If channels leave the protected set as decode position increases, then a policy calibrated early in a trace can be stale later in the same trace.

The deployment pressure is not abstract. Artificial Analysis benchmark-cost data reported by Bort (2025) put OpenAI o1 at $2,767.05 to evaluate across seven reasoning-heavy benchmarks, compared with $108.85 for GPT-4o on the same suite; the report attributes the roughly 25× cost gap to both token expansion and higher per-token prices. Gartner's March 2026 inference-cost forecast predicts that trillion-parameter LLM inference will become over 90% cheaper by 2030 than in 2025, while warning that advanced reasoning demand can still make total compute scarce (Gartner, 2026). Commercial market forecasts are not evidence about model internals, but they indicate scale: one MarketsandMarkets report projects the AI inference market from $106.15B in 2025 to $254.98B in 2030, and another projects edge AI hardware from $26.14B in 2025 to $58.90B in 2030 (MarketsandMarkets, 2025a;b). Long-decode quantization failures therefore matter both for datacenter reasoning workloads and for eventual edge reasoning deployments.

The current status is a measurement and mechanism paper. The empirical result has become stronger than the original Granite/Mamba-specific framing: decode-position channel drift now appears in Granite hybrid models, a Nemotron-3 hybrid, a pure-Transformer DeepSeek control, and Falcon-H1's parallel hybrid topology. The method result is weaker: three static or lightly position-conditioned W4A16 interventions and three later Phase 9 policies fail to produce a reliable protection method. This is not a positive systems paper yet. Its value is that it identifies three candidate mechanisms sharply enough to constrain future methods: harmful boundary discontinuities, smooth-but- insufficient set updates, cross-tensor coupling that is not arbitrary but still insufficient, and the remaining split between reactive selection, insufficient budget, and compound error.

The contribution is:

- quantitative characterization of decode-position channel-set drift at reasoning-scale horizons (up to 20,000 decode tokens) across hybrid Mamba-2 and pure-Transformer reasoning LLMs, extending DecDEC's short-horizon Transformer evidence to the long-decode reasoning regime while contradicting the channel-persistence assumption that motivates Quamba2-style static ordering;
- cross-architecture replication on Nemotron-3, DeepSeek-R1-Distill, and Falcon-H1 under fixed deterministic traces;
- strict set-membership decomposition showing that many channels leave the protected top-1% set entirely, rather than merely shuffling within it;
- negative intervention evidence showing that static union protection, larger scoring windows, simple position-bin switching, position-binned scale tables, EMA-smoothed set updates, activation/KV coupling, and a DecDEC-style reactive baseline do not yet yield a positive W4A16 protection method on the Granite-Small testbed.

The paper does not claim a deployable quantization recipe, a kernel/runtime system, a performance improvement, or completed architecture-general validation.

2 Related Work

Mamba quantization and dynamic outliers. OutlierMigrate is not the first work to observe dynamic outlier behavior in Mamba-style models. In vision state-space models, QMamba reports that hidden state sequences exhibit highly dynamic variations and introduces Temporal Group Quantization for hidden states (Li et al., 2025). OuroMamba makes the point even more directly for vision Mamba models: it identifies dynamic activation-outlier variation across time steps as a failure mode for static post-training quantization and uses lightweight dynamic outlier detection during inference (Ramachandran et al., 2025). These results are important prior art. They establish that dynamic outliers are already known in the vision-Mamba setting.[1]

[1] The phrase "outlier migration" is overloaded. In this paper, the operational measurement is decode-position channel drift: which activation channels are in the top-magnitude set at different decode positions. SmoothQuant uses migration for activation-to-weight difficulty transfer (Xiao et al., 2023). MoBiQuant uses it for token-set shifts under different bit widths (Wang et al., 2026); we reserve our quantitative claims for the channel-across-decode-position phenomenon measured in this paper.

Language-Mamba quantization work has emphasized a different operating regime. Mamba-PTQ shows that recurrent LLMs exhibit activation outlier channels and frames outlier-aware quantization as a first step (Pierro and Abreu, 2024). Quamba proposes a static 8-bit per-tensor recipe that suppresses maximum input values to selective SSMs and uses a Hadamard transform for output activations (Chiang et al., 2024). Quamba2 is closer to the assumption tested here: it explicitly builds on channel-order preservation and activation persistence for Mamba1/Mamba2 quantization (Chiang et al., 2025). Our contribution is not that dynamic channel sets exist in every setting; it is that the Quamba2-style persistence assumption does not hold for the measured reasoning-scale W4A16 decode horizons. MambaQuant uses variance-aligned rotations and reports heavy-tailed, channel-uneven distributions in Mamba-family models (Xu et al., 2025). Quamba-SE softens the treatment of activation outliers with multiple adaptive scales but still evaluates a per-value quantizer rather than a decode-time channel migration policy (Chen and Hemani, 2026). OutlierMigrate asks which regime applies to hybrid LLMs in long reasoning traces: the static or calibration-time protection regime common in language deployment recipes, or the dynamic regime already documented in vision Mamba.

Static protection in LLM deployment. Several influential LLM quantization systems protect or transform salient activation structure using calibration-time statistics or fixed transformations. SmoothQuant migrates activation difficulty into weights using offline activation statistics (Xiao et al., 2023); AWQ protects a small set of activation-salient weight channels (Lin et al., 2023); QuaRot removes hidden state outliers with rotations (Ashkboos et al., 2024); KVQuant isolates KV cache outliers per vector and uses per-channel key quantization (Hooper et al., 2024); and BlockDialect assigns block-wise number formats from a learned formatbook (Jang and Tambe, 2025). These systems are not claimed to be wrong by our current evidence. The implication is narrower: if the protected high-magnitude set changes substantially during long decode in hybrid LLMs, static protection maps may need migration-aware updates or validation on the specific recurrent/linear-attention architecture. Several of these systems already include adaptive components, such as KVQuant's per-vector dense-and-sparse handling or BlockDialect's online activation format selection; our current evidence motivates stress-testing those designs on hybrid long decode, not declaring them failed.

Dynamic channel and KV-cache policies. Dynamic inference-time policies are closer to the methods we test and fail. DecDEC dynamically selects salient activation channels at each decode step and fetches residual compensation from CPU memory (Park et al., 2025). DecDEC also reports low recall for static outlier-channel identification at short decode horizons. Our measurement is therefore not a first demonstration that channel sets change during decode; the defensible distinction is horizon, model class, architecture, and benchmark surface: up to 20,000 decode tokens on reasoning LLMs, including hybrid Mamba-2 and parallel-hybrid models, rather than DecDEC's short-horizon Transformer setting. M2 and hard-binned M10 differ from DecDEC in that they switch a precomputed protected-channel or scale policy by decode-position bin; M11 tests whether smoothing that decision surface avoids the boundary discontinuities exposed by M2 and M10. It does avoid the random-control-beats pattern but still does not recover enough quality. KV-cache systems address a different tensor. KIVI quantizes key cache per channel and value cache per token (Liu et al., 2024); PM-KVQ targets long-CoT KV-cache quantization and RoPE-induced key-cache distribution shift (Liu et al., 2025); and AttentionPredictor learns to predict future attention scores for KV-cache compression and prefetch (Yang et al., 2025). These works motivate long-decode inference adaptation, but our measurements are about linear-layer activation channel sets.[2] PMPD changes precision across inference phases and deeper decode positions (Chen et al., 2025); our current evidence points in an orthogonal direction,

[2] "Hot channels" in HCP/CHON (arXiv:2602.02047) refers to persistently extreme channels during NVFP4 pretraining, which is close in vocabulary but opposite in setting from the drifting W4A16 inference-time channels measured here.

where later decode positions can require different protected channels rather than simply lower precision.[3]

Channel-wise gated recurrence. Recent efficient-recurrence and linear-attention architectures make data-dependent channel-wise memory control a central design axis. Gated Linear Attention (GLA) formulates linear attention as recurrent matrix-valued state with data-dependent gating (Yang et al., 2023). Gated DeltaNet combines adaptive memory erasure with delta-rule updates and reports hybrids that mix Gated DeltaNet with attention or Mamba2 layers (Yang et al., 2024). Moonshot AI's October 2025 Kimi Linear report is a recent example in this line: its Kimi Delta Attention (KDA) is described as channel-wise gated linear attention that extends Gated DeltaNet with finer-grained gating, and the released Kimi Linear model is a layerwise hybrid of KDA and Multi-Head Latent Attention (Moonshot AI Kimi Team, 2025). The official Qwen3.6 model card likewise describes a hybrid layout that interleaves Gated DeltaNet blocks with gated attention (Qwen Team, 2026). These sources motivate OutlierMigrate as an empirical test of whether high-magnitude channels stay fixed during long decode in hybrid language models. They do not imply that Kimi Linear, Qwen3.6, GLA, or other unmeasured models exhibit the migration measured in this draft.

3 Method

Phase 0 follows the frozen preregistration at experimental/outlier_migrate/phase0/preregister_om_phase0.md. The model is ibm-granite/granite-4.0-h-tiny at HuggingFace snapshot commit 791e0d3d28c86e106c9b6e0b4cecdee0375b6124. The prompt set contains 12 deterministically selected AIME-2025 traces, indices 0–11, with prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json.

Phase 1 follows the frozen preregistration at experimental/outlier_migrate/phase1/preregister_om_phase1.md. The model is ibm-granite/granite-4.0-h-small at HuggingFace snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39. The prompt set contains 24 deterministically selected AIME-2025 traces, indices 0–23, with prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json.

For every transformer layer, the runner records per-channel activation magnitudes. Phase 0 records decode positions {100, 500, 1000, 5000, 10000}; Phase 1 records {100, 500, 1000, 5000, 10000, 20000}. Each gate selects the top 1% channels by mean magnitude at position 100 and computes the fraction whose rank at the final preregistered position differs by more than two channel positions. The reported gate metric is the trace-bootstrap aggregate over layers with equal layer weighting.

4 Experiments

The Phase 0 result packet is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z. The checker output is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/artifact_check.json. The activation artifact SHA is sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687 from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json.

[3] We refer to arXiv:2603.19296 as test-time TTQ when discussing it, to avoid confusion with classical Trained Ternary Quantization.

The Phase 1 result packet is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z. The checker output is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/artifact_check.json. The activation artifact SHA is sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1 from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

The partial Phase 2 result packet is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. The checker output is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/artifact_check.json. The model is nvidia/NVIDIA-Nemotron-3-Nano-30B-A3B-BF16 at snapshot commit cbd3fa9f933d55ef16a84236559f4ee2a0526848. The activation artifact SHA is sha256:bc841b938a125f05649f84577c0ddb7ab287213a683892cb81ea6ca06296206d from experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z/activation_magnitude_manifest.json. This packet is explicitly scoped as partial Nemotron-3-only validation: Qwen3.6 and Kimi Linear are deferred, and full_cross_validation_complete=false.

The Phase 3 intervention packet is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z. It follows the frozen preregistration at experimental/outlier_migrate/phase3/preregister_om_phase3_intervention.md. The checker output is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/checker_result.json, the metrics file is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json, the per-trace perplexities and recoveries are in experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json, and artifact completeness is recorded in experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/artifact_check.json. The model is again ibm-granite/granite-4.0-h-tiny at snapshot commit 791e0d3d28c86e106c9b6e0b4cecdee0375b6124. The prompt set uses the fixed 24-trace Phase 1 set with prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e. The quantization scheme is simple symmetric per-channel INT4 weights with FP16 activations and BF16 protected-channel rows; no AWQ-style or SmoothQuant-style activation-aware scaling is used. The scored window is the 64-token window ending at decode position 10000. The preregistered recovery metric is

$$1 - \frac{\text{perplexity}_{\text{regime}} - \text{perplexity}_{\text{BF16}}}{\text{perplexity}_{\text{static1\%}} - \text{perplexity}_{\text{BF16}}}.$$

The pass rule is median recovery ≥ 0.50 with bootstrap 95% CI lower bound > 0.30. The kill rule is recovery < 0.20 or CI upper < 0.30. The mandatory controls are static position-100 top-2% protection and magnitude-average top-1% protection over positions $\{100, 1000, 5000, 10000\}$.

The preregistered dynamic-outlier pass rule in both gates requires migration fraction at least 0.05 and bootstrap CI lower bound greater than 0.05. This is a nonstationarity screen, not a claimed practically meaningful effect-size standard. Phase 1 uses the same dynamic path because Phase 0 returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION.

5 Results

The original preregistered migration fraction intentionally conflates two effects: channels can leave the top-1% set entirely, or they can remain in the set while shuffling rank by more than two positions. For static channel protection, the stricter set-membership readout is

Quantity	Phase 0	Phase 1
Decision	dynamic-outlier pass	replicated-at-scale pass
Strict set-leaving fraction	0.634244791667	0.566234756098
Strict set-leaving 95% CI	[0.605208333333, 0.664192708333]	[0.550076219512, 0.581707317073]
Within-set rank-shuffling fraction	0.175260416667	0.270934959350
Within-set rank-shuffling 95% CI	[0.160156250000, 0.191015625000]	[0.260797764228, 0.280614837398]
Original preregistered migration fraction	0.8178385416666667	0.843165650406504
Original migration 95% CI	[0.797265625, 0.8368489583333334]	[0.8334349593495936, 0.8511432926829268]
Preregistered threshold	≥0.05 and CI lower >0.05	≥0.05 and CI lower >0.05
Artifact complete	true	true
Model	ibm-granite/granite-4.0-h-tiny	ibm-granite/granite-4.0-h-small
Trace count	12	24
Layer count / hidden size	40 / 1536	40 / 4096
Decode positions	100, 500, 1000, 5000, 10000	100, 500, 1000, 5000, 10000, 20000
Bootstrap seed	20260508	20260508

Table 1: OutlierMigrate Phase 0 and Phase 1 gate results. Phase 0 values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/checker_result.json and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/migration_decomposition.md. Phase 1 values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/checker_result.json and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/metrics.json; strict decomposition values are from experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/migration_decomposition.md.

the more direct post-hoc interpretability measure, not the preregistered gate criterion. A channel is counted as strict set-leaving only if it is in the per-trace, per-layer top-1% set at decode position 100 and has final rank greater than the top-1% boundary at decode position 10000 in Phase 0 or 20000 in Phase 1. Under that definition, 0.634244791667 of Phase 0 base top-1% channels and 0.566234756098 of Phase 1 base top-1% channels leave the protected set entirely. The within-set shuffling rates are 0.175260416667 and 0.270934959350, respectively. The gate decision remains based only on the original preregistered migration metric.

The exact Phase 0 provenance values are: model snapshot commit 791e0d3d28c86e106c9b6e0b4cecdee0375b6124, prompt SHA sha256:2f27c54baa8448e033d6e82f53f775dc6abe38188e4f1e5c0b97e3c74fe7c1dd, and activation artifact SHA sha256:2783aa329d82e8f43ea4f342e52f0a25fc283b0b8971880f2da093739474d687; these are recorded in experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/model_provenance.json, experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/prompt_manifest.json, and experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z/activation_magnitude_manifest.json. The exact Phase 1 provenance values are: model snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39, prompt SHA sha256:aa038b29332b6d137d558205ee441163e7ea4cb3cc323eb705a2f5928fd2fe4e, and activation artifact SHA sha256:326a04351e0dda28cf919fe4745d7d9341011db13d940a06cf9c8470633003e1; these are recorded in experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/model_provenance.json, experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/prompt_manifest.json, and experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z/activation_magnitude_manifest.json.

Quantity	Partial Phase 2 Nemotron-3
Decision	partial PASS; Qwen3.6/Kimi deferred
Strict set-leaving fraction	0.533713200380
Strict set-leaving 95% CI	[0.467414529915, 0.599982193732]
Within-set rank-shuffling fraction	0.269082383666
Within-set rank-shuffling 95% CI	[0.238841405508, 0.299056267806]
Original preregistered migration fraction	0.820809591642925
Original migration 95% CI	[0.7865325261158594, 0.8544931149097815]
Preregistered threshold	$\geq$0.05 and CI lower >0.05
Artifact complete	true
Model	nvidia/NVIDIA-Nemotron-3-Nano-30B -A3B-BF16
Trace count	24
Layer count / hidden size	52 / 2688
Decode positions	100, 500, 1000, 5000, 10000, 20000

Table 2: Partial Phase 2 cross-family validation. The checker-backed packet is experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. This is not full cross-validation because Qwen3.6 and Kimi Linear are deferred.

Run	Original drift	Strict set-leaving	Within-set shuffling	Packet
Granite-Small Phase 1	0.836763211382	0.566234756098	0.270934959350	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
Nemotron-3 Phase 2	0.801667853751	0.533713200380	0.269082383666	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
DeepSeek-R1-Distill Phase 5'	0.834914434524	0.670572916667	0.165736607143	experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md
Falcon-H1 Phase 7	0.767887205387	0.673611111111	0.097537878788	experimental/outlier_migrate/phase7/results/om_phase7_falcon_h1_20260512T223600Z/checker_result.json

Table 3: Cross-architecture decomposition after Phase 5' and Phase 7. Strict set-leaving remains above 0.50 on Granite-Small, Nemotron-3, DeepSeek, and Falcon-H1. The Falcon-H1 packet uses post-block residual hooks; it is not pathway-specific pre-sum attribution.

The Phase 0 checker returned PASS_OM_PHASE0_DECODE_TIME_MIGRATION. Its exact artifact reason was that the point estimate 0.81783854 is at least 0.05 and the CI lower bound 0.79726562 is greater than 0.05. The Phase 1 checker returned PASS_OM_PHASE1_REPLICATED_AT_SCALE. Its exact artifact reason was that the point estimate 0.84316565 is at least 0.05 and the CI lower bound 0.83343496 is greater than 0.05.

Phase 5' broadens the phenomenon beyond measured Mamba-2-style hybrids: deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B returns DYNAMIC_REGIME_TRANSFORMER with original drift 0.8393787202380952, 95% CI [0.8278459821428571, 0.8507254464285714], strict set-leaving 0.670572916667, and within-set shuffling 0.165736607143. Phase 7 checks Falcon-H1 as a parallel hybrid topology and returns WITHIN_LINEAGE_2_CONSISTENT: original drift 0.7833543771043772, 95% CI [0.7544191919191919, 0.8112373737373737], strict set-leaving 0.673611111111, and within-set shuffling 0.097537878788. These two additions make the Mamba-specific title untenable and motivate the broader "decode-position channel drift" framing.

Experiment D formalizes the decomposition and Experiment E checks threshold sensitivity. The top-1% decomposition is stable through top-2% on all landed packets. Top-5% changes the component split and is treated as sensitivity, not the primary operating point. Phase 9 Step 9.0 additionally computes adjacent-bin Jaccard overlap for top-channel sets; the overall top-1% adjacent-bin Jaccard is 0.745999647119, which made position-binned methods plausible enough to test.

5.1 Static and position-conditioned protection attempts

Phase 3 tests the most direct intervention suggested by the measurement: instead of protecting only the position-100 top-1% channels, protect the union of top-1% channels observed at decode positions {100, 1000, 5000, 10000}. This is intentionally a static-set intervention. It asks whether membership migration alone explains enough of the quantization gap to support a simple deployment recipe.

Regime	Mean perplexity	Median perplexity	Median recovery	Recovery 95% CI
BF16 baseline	1.005358821083984	1.000282908378305	reference	reference
static-1% position 100	1.015228246226743	1.000282342709987	reference	reference
static-2% position 100	1.021467448964073	1.000390886047853	0.000000000000	[0.000000000000, 0.356017001423]
magnitude average top-1%	1.009248547916613	1.000236584155715	0.061276118929	[0.000000000000, 0.512245438462]
migration-aware union	1.008573512406127	1.000275313560728	0.000000000000	[0.000000000000, 0.711143244199]

Table 4: Phase 3 intervention and controls on Granite-4.0-H-Tiny. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/per_trace_metrics.json, experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/metrics.json, and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/control_metrics.json. Recovery is defined relative to the BF16-vs-static-1% gap.

Grid	Positions	Median recovery	Recovery 95% CI
sparse	{100, 5000, 10000}	0.000000000000	[0.000000000000, 0.752281234025]
primary	{100, 1000, 5000, 10000}	0.000000000000	[0.000000000000, 0.711143244199]
dense	{100, 500, 1000, 2000, 5000, 7500, 10000}	0.000000000000	[0.000000000000, 0.920007799726]

Table 5: Position-grid sensitivity. Values are from experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity.md and experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z/grid_sensitivity_metrics.json. Increasing grid density did not produce a nonzero median recovery.

The intervention kills. The checker returned the decision string KILL_OM_PHASE3_INTERVENTION_FAILS because the union regime had median recovery 0.000000000000, below the preregistered 0.20 kill floor. It also missed the pass rule: median recovery ≥ 0.50 with CI lower bound > 0.30. The artifact packet is complete, and the control-stop condition did not fire. The best control, magnitude-average protection, exceeded union by 0.061276118929 median recovery, below the human stop threshold of 0.10. Ten of the 24 traces had no recoverable static gap under this scoring definition. That fraction, 0.416666666667, also exceeds the preregistered 25% no-gap kill condition. The checker reports the first kill reason it reaches, but the packet satisfies both kill paths.

The recovery statistic is intentionally centered on the per-trace BF16-vs-static-1% gap, so it can be negative when a regime is worse than static-1%, greater than one when a regime beats BF16 on the scored window, and zero when there is no recoverable static gap. In the union regime, 11 traces had positive recovery, 10 were zero because there was no recoverable static gap, 3 were negative, and 4 exceeded recovery 1.0. This unstable per-trace surface is part of the negative result rather than a hidden success case: the primary statistic was preregistered as the median across all 24 traces. The median was chosen for the gate because the recovery ratio can become very large when the static gap is small; a mean recovery would be dominated by unstable high-ratio traces rather than by the typical trace.

Phase 4 repeats the union intervention on Granite-4.0-H-Small with a larger 512-token scoring window. It also kills: median recovery is 0.000000000000, CI95 [0.000000000000, 0.06954064195470389], and the no-recoverable-static-gap fraction is 0.375. This rules out the simplest explanation that Phase 3 failed only because Granite-Tiny was too robust or because the 64-token scoring window was too short.

Phase 9 M2 then tests a lightly position-conditioned intervention. Instead of one static union, M2 uses three protected sets: early positions {100, 200, 500}, middle positions {1000, 2000, 5000}, and late positions {7000, 10000, 15000}. The active protected set switches at decode positions 800 and 6000. The result kills by the random-control rule. On the vacation-mode 12-trace deterministic slice, 8 traces have a positive static-1% gap. M2 median recovery is -0.866837313391, static-3% matched-cost median recovery is -0.625927638959, and random-bin median recovery is -0.199289023223. The random-bin assignment beats M2 by 0.667548290168 median recovery, exceeding the preregistered 0.10 kill threshold. The random-bin control is not good in absolute terms; its importance is that it is less bad than the intended position-conditioned assignment.

Attempt	Model	Decision	Median recovery	CI95	Key control readout
Phase 3 union	Granite-Tiny	KILL_OM_PHASE3_INTERVENTION_FAILS	0.000000000000	[0.000000000000, 0.711143244199]	10/24 no recoverable static gap
Phase 4 union	Granite-Small	KILL_OM_PHASE4_INTERVENTION_FAILS	0.000000000000	[0.000000000000, 0.069540641955]	9/24 no recoverable static gap
Phase 9 M2 bins	Granite-Small	KILL_M2_RANDOM_CONTROL_BEATS	-0.866837313391	[-3.435289251335, 0.595238665766]	random-bin median beats M2 by 0.667548290168
Phase 9 M10 scales	Granite-Small	KILL_M10_RANDOM_CONTROL_BEATS	0.234447927834	[-2.072905653259, 0.503521494450]	random-bin median beats M10 by 0.761487459046
Phase 9 M11 EMA	Granite-Small	KILL_M11_AMBIGUOUS	0.048299284138	[-8.941156151014, 0.704599255873]	random-walk median is lower by 1.254404832487
Phase 9 M18 act+K	Granite-Small	KILL_M18_AMBIGUOUS	-0.343590844126	[-14.071191710979, 0.740602124280]	KIVI key-only median -2.174128311393; random-coupled median -4.664379731051
Phase 9 DecDEC proxy	Granite-Small	PASS_DECDEC_BASELINE_REPORTED	-0.070034781258	[-8.495693501233, 0.674176370350]	random reactive median -1.262604926125; static top-10% median -0.228273368210

Table 6: Intervention attempts. Phase 3 and Phase 4 test static union protection under W4A16. Phase 9 M2 tests position-conditioned protected sets A/B/C with a static-3% matched-cost control and a random-bin negative control. Phase 9 M10 tests hard-binned scale tables. Phase 9 M11 tests EMA-smoothed protected-set updates. Phase 9 M18 tests joint activation and key-cache channel protection. The DecDEC row is a descriptive algorithmic baseline, not a positive-method gate. None provides a positive method. M10 values are from experimental/outlier_migrate/phase9/results/om_phase9_m10_granite_small_vac12_20260515T085800Z/checker_result.json; M11 values are from experimental/outlier_migrate/phase9/results/om_phase9_m11_granite_small_vac12_20260516T010728Z/checker_result.json; M18 values are from experimental/outlier_migrate/phase9/results/om_phase9_m18_granite_small_vac12_20260516T193500Z/checker_result.json; DecDEC values are from experimental/outlier_migrate/phase9/results/om_phase9_decdec_granite_small_vac12_20260517T141500Z/checker_result.json.

M10 tests whether the problem is not membership but scale choice: it stores separate SmoothQuant-style scale tables for decode-position bins. This also kills by the random-control rule. M10 median recovery is 0.234447927834, CI95 [-2.072905653258708, 0.5035214944499808], while the random-bin scale control has median recovery 0.995935386880 and beats M10 by 0.761487459046. M11 then removes the hard boundary by replacing bin switches with an EMA-smoothed protected-set score. M11 does not pass: the best arm, alpha 0.5, has median recovery 0.048299284138 with CI95 [-8.94115615101423, 0.7045992558730417]. However, its random-walk control has median recovery -1.206105548349, so M11 is structurally different from M2 and M10: smoothing removes the random-control-beats failure, but smoothing alone does not recover quality.

Together, the intervention results make a useful negative statement with three mechanisms. First, boundary discontinuities are actively harmful: M2 and M10 are both beaten by random-bin controls. Second, smoothing removes that harm but does not solve the gap: M11 is not beaten by random-walk, yet misses the positive-method bar. Third, simple activation/KV coupling has relative signal but is still insufficient: M18 activation+K has median recovery -0.343590844126, while KIVI key-only and random coupled controls have medians -2.174128311393 and -4.664379731051. That comparison suggests the coupled selection is not arbitrary noise, but it does not recover quality. The DecDEC algorithmic proxy then tests a reactive endpoint-selection baseline: median recovery is -0.070034781258 with CI95 [-8.495693501233, 0.674176370350]. It is less damaging than a random reactive top-1% control whose median is -1.262604926125, but it does not recover the W4A16 static gap. The remaining open question is whether a larger protected-channel budget, stable-core targeting, or long-horizon compound error better explains the gap. M11b, M26, and the KL accumulation analysis are the next mechanism tests.

A post-M18 diagnostic pass further narrows what the current packets can say. The K/V-cache migration question is not identifiable from existing artifacts: only M18 contains K/V-channel evidence, and that evidence is endpoint-only at decode position 10000 rather than early-to-late K/V snapshots. The recovery curve question is also not identifiable, because M2, M10, M11, and M18 each score one 512-token endpoint window. The activation-side evidence is still useful: Granite-Small fine-grid top-1% Jaccard changes smoothly across 500-position bins, and all measured models retain a small always-protected core that is smaller than the full top-1% set. These diagnostics make DecDEC and M11b the right next tests rather than another unmotivated method: DecDEC tests reactive endpoint selection, while M11b tests whether the remaining failure is simply protection-budget insufficiency.

Run / layer type	Layers	Strict set-leaving	Within-set shuffling
Phase 0 Granite-Tiny attention	4	0.618490	0.186198
Phase 0 Granite-Tiny SSM/Mamba	36	0.635995	0.174045
Phase 1 Granite-Small attention	4	0.557165	0.282266
Phase 1 Granite-Small SSM/Mamba	36	0.567243	0.269676
Phase 2 Nemotron-3 attention	6	0.563014	0.254630
Phase 2 Nemotron-3 MoE	23	0.526503	0.277778
Phase 2 Nemotron-3 SSM/Mamba	23	0.533280	0.264157

Figure 1: Layer-stratified migration decomposition from experimental/outlier_migrate/phase3/results/layer_stratified_migration.json. Values report layer-type means, averaged equally over layers. Strict set-leaving is the portion most directly relevant to static protected-channel membership; within-set shuffling motivates scale-refresh mechanisms but is not itself a membership failure.

5.2 Layer-stratified migration

The negative intervention result does not erase the measurement finding: it narrows the mechanism. Figure 1 uses the existing Phase 0, Phase 1, and partial Phase 2 activation packets to stratify strict set-leaving and within-set shuffling by layer type. The result is not confined to a single Granite mixer family. Granite shows strict set-leaving in both attention and SSM/Mamba layers, while Nemotron-3 shows it in attention, SSM/Mamba, and MoE-classified layers. This is consistent with the paper's diagnostic claim: static membership is unstable in the measured packets, but successful handling likely needs more than a larger static protected set.

6 Discussion

The strongest claim in this draft is now broader and more precise than the initial Granite-specific hypothesis. Across four measured model families, channels that are top-1% by magnitude early in a long decode often do not remain in the later top-1% set. The strict set-leaving decomposition is the most important systems readout because it directly targets the membership assumption behind static protected-channel maps. The original rank-drift metric remains the preregistered gate; the stricter decomposition explains why the gate matters for deployment.

The negative method results are also informative. Static union protection addresses only set membership; it does not refresh scales, respond to within-set rank changes, or decide which layers need different treatment. Phase 4 shows that the Phase 3 failure is not just a Granite-Tiny or 64-token window artifact. Phase 9 M2 shows that hard protected-set binning can be actively harmful, and M10 shows the same pattern for hard scale-table binning. M11 then separates smoothness from recovery: it removes the random-control failure but does not restore quality. The mechanism lesson is therefore sharper than "use a bigger protected set" or "smooth the set." Long-decode channel drift appears real, but simple static, hard-binned, and activation-only smoothed policies are inadequate.

This leaves three plausible future directions. First, a reactive oracle or learned predictor may be required because current channel identity is stale for future importance; M18's negative result weakens the simplest activation/KV-coupling version of that hypothesis without eliminating the reactive-selection hypothesis. Second, the budget may be too small: a smooth policy could need top-5% or top-10% protection before recovery appears. Third, per-step protection may be attacking the wrong surface if the dominant loss is compound error accumulated over thousands of decode steps. The active Phase 1 queue is now ordered around the two remaining discriminators: DecDEC supplies a reactive per-step baseline, and M11b tests whether larger smoothed budgets recover the gap.

The theoretical mechanism remains plausible but is not itself an empirical claim about unmeasured models. In channel-wise gated recurrence or linear-attention layers, independent per-channel decay rates imply that channels retain state at different rates. Low-decay channels can accumulate magnitude over long decode, while high-decay channels shed magnitude. Under heterogeneous inputs and heterogeneous decay rates, the composition of the top 1% channels by magnitude should change across decode positions rather than remain fixed to the position-100 set. Kimi Linear's KDA, Qwen3.6-style Gated DeltaNet blocks, and GLA-style gated recurrence provide architectural examples where channel-wise or fine-grained data-dependent gating is central (Moonshot AI Kimi Team, 2025; Qwen Team, 2026; Yang et al., 2024; 2023). This draft has not measured Kimi Linear, Qwen3.6, or GLA variants.

7 Limitations

This draft is not camera-ready final and should not be read as a positive systems method. It reports replicated measurement and failed method attempts, not an end-to-end efficiency or quality improvement. The strongest positive evidence is the cross-model measurement; the strongest negative evidence is that three simple W4A16 protection policies fail under their frozen gates.

The scope is W4A16 post-training quantization for small reasoning models and hybrid Mamba-2 or parallel-hybrid architectures, with a small pure-Transformer reference. It should not be read as an all-formats claim. The underlying decode-position channel-set drift may be a broader autoregressive phenomenon, but quality consequences depend on format dynamic range: INT4 exposes severe failure surfaces here, FP8 may absorb comparable drift in settings like ThoughtFlow, block formats such as MXFP4/NVFP4 move the analysis from channels to blocks, and INT2/INT3 would likely be still more fragile. Large standard reasoning models are also outside the current evidence. Recent reasoning PTQ work such as ParoQuant reports strong W4A16 recovery on larger reasoning models (Liang et al., 2025); the required ParoQuant baseline for this draft is queued but not yet run.

The empirical surface is still limited. All reported traces are deterministic AIME-2025 slices. The bootstrap uncertainty is over traces, not over independent benchmark families, seeds, or prompt distributions. The top-1% fraction, rank-delta threshold, and decode-position grid were fixed in the packets; they should not be retuned on these results. Experiment E shows reasonable top-1% to top-2% stability, but top-5% changes the component split.

Cross-lineage validation remains incomplete. Nemotron-3, DeepSeek, and Falcon-H1 reduce Granite-specific risk, but Qwen3.6 was blocked by hook accessibility in current SGLang infrastructure without source modification, and Kimi Linear has not been measured. Falcon-H1 is post-block residual evidence rather than separate pre-sum SSM/Attention pathway attribution. Generalization to RWKV-7, GLA-style systems, and other efficient-recurrence variants is not established.

Finally, the method failures do not prove that decode-position channel drift is unusable. They prove that static union sets, a larger Granite-Small scoring window, simple hard-binned policies, one EMA-smoothed activation-only policy, and one activation/KV coupling policy are not enough under this quantization setup. The next credible method evidence should come from the remaining mechanism-distinguishing Phase 1 queue: DecDEC algorithmic baseline and M11b budget scaling, not from retuning failed thresholds.

8 Reproducibility Statement

The Phase 0 result packet is `experimental/outlier_migrate/phase0/results/om_phase0_20260508T011824Z`. It includes environment.json, model_provenance.json, prompt_manifest.json, command_metadata.json, random_seed.json, metrics.json, bootstrap_ci.json, checker_result.json, and artifact_check.json. The recorded Phase 0 command was:

experimental/shared/run_phase0_branch.py --branch outlier_migrate

The Phase 1 result packet is experimental/outlier_migrate/phase1/results/om_phase1_20260508T014959Z. It includes the same artifact classes and records the Granite-4.0-H-Small snapshot commit b8c0982bab7fde4eb48110f5a069527c008fab39. The recorded Phase 1 command was:

experimental/outlier_migrate/phase1/run_om_phase1.py

The checker decisions and exact numbers in this draft are taken from the two Granite result directories above and the partial Phase 2 directory experimental/outlier_migrate/phase2/results/om_phase2_nemotron3_20260508T231723Z. The OutlierMigrate runners emit the artifact-checked metric file metrics.json; they do not emit a separate profiler_metrics.json. The recorded Phase 2 command was:

```
experimental/outlier_migrate/phase2/run_cross_model.py \
  --partial-nemotron3-only \
  --run-id om_phase2_nemotron3_20260508T231723Z \
  --batch-size 4 --dtype bfloat16 --device auto
```

The Phase 3 intervention packet is experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z. It includes environment.json, model_provenance.json, prompt_manifest.json, command_metadata.json, random_seed.json, per_trace_metrics.json, metrics.json, control_metrics.json, grid_sensitivity_metrics.json, checker_result.json, and artifact_check.json. The recorded Phase 3 command was:

```
experimental/outlier_migrate/phase3/run_phase3_intervention.py \
  --run-id om_phase3_20260509T212000Z \
  --batch-size 4 --dtype bfloat16 --device auto \
  --reuse-prequant-run-dir \
  experimental/outlier_migrate/phase3/results/om_phase3_20260509T180200Z
```

The grid-sensitivity artifact was generated with:

```
experimental/outlier_migrate/phase3/run_grid_sensitivity.py \
  --run-dir \
  experimental/outlier_migrate/phase3/results/om_phase3_20260509T212000Z
```

Figure 1 is a LaTeX-native summary table generated from experimental/outlier_migrate/phase3/results/layer_stratified_migration.json.

The Phase 4 intervention packet is experimental/outlier_migrate/phase4/results/om_phase4_20260511T054000Z. The Phase 5' Transformer-control packet is experimental/outlier_migrate/phase5_prime/results/om_phase5p_20260512T053800Z. Experiment D and E artifacts are in experimental/outlier_migrate/decomposition_analysis/. The Phase 7 Falcon-H1 packet is experimental/outlier_migrate/phase7/results/om_phase7_falcon_h1_20260512T223600Z. Phase 9 Step 9.0 outputs are experimental/outlier_migrate/phase9/step9_0_decomposition_replication.md and experimental/outlier_migrate/phase9/step9_0_bin_overlap_analysis.md. The Phase 9 M2 packet is experimental/outlier_migrate/phase9/results/om_phase9_m2_granite_small_vac12_finalized_20260514T233800Z. Its checker decision is KILL_M2_RANDOM_CONTROL_BEATS; the vacation-mode adaptation notes are in swarm/vacation_decisions/. The Phase 9 M10 packet is experimental/outlier_migrate/phase9/results/om_phase9_m10_granite_small_vac12_20260515T085800Z; the Phase 9 M11 packet is experimental/outlier_migrate/phase9/results/om_phase9_m11_granite_small_vac12_20260516T010728Z; and the Phase 9 M18 packet is experimental/outlier_migrate/phase9/results/om_phase9_m18_granite_small_vac12_20260516T193500Z. Their checker decisions are KILL_M10_RANDOM_CONTROL_BEATS, KILL_M11_AMBIGUOUS, and KILL_M18_AMBIGUOUS. Post-M18 analysis reports are in experimental/outlier _migrate/phase9/post_m18_analysis/; they document which mechanism questions are identifiable from the landed packets and which would require new telemetry.

9 Ethics Statement

This is an activation-analysis measurement study on benchmark prompts. The current artifact does not make deployment, user-data, safety, systems-efficiency, or capability-improvement claims. The main ethical risk is overclaiming the replicated measurement or the failed Phase 3/4/9 interventions as a validated cross-model positive method; this draft explicitly rejects that interpretation until a future intervention pass.

References

Julie Bort. The rise of AI reasoning models is making benchmarking more expensive. TechCrunch, April 10 2025. https://techcrunch.com/2025/04/10/the-rise-of-ai-reasoning-models-is-making-benchmarking-more-expensive/.

Gartner. Gartner predicts that by 2030, performing inference on an LLM with 1 trillion parameters will cost GenAI providers over 90% less than in 2025. Press release, March 25 2026. https://www.gartner.com/en/newsroom/press-releases/2026-03-25-gartner-predicts-that-by-2030-performing-inference-on-an-llm-with-1-trillion-parameters-will-cost-genai-providers-over-90-percent-less-than-in-2025.

MarketsandMarkets. AI inference market size, share and growth, 2025 to 2030. Commercial market report, February 2025. https://www.marketsandmarkets.com/Market-Reports/ai-inference-market-189921964.html.

MarketsandMarkets. Edge AI hardware industry worth $58.90 billion by 2030. Commercial market report press release, July 4 2025. https://www.marketsandmarkets.com/PressReleases/edge-ai-hardware.asp.

Dongwei Wang, Jinhee Kim, Seokho Han, Denis Gudovskiy, Yohei Nakata, Tomoyuki Okuno, KhayTze Peong, Kang Eun Jeon, Jong Hwan Ko, Yiran Chen, and Huanrui Yang. MoBiQuant: Mixture-of-Bits Quantization for Token-Adaptive Elastic LLMs. arXiv:2602.20191, February 2026. https://arxiv.org/abs/2602.20191.

Yeonhong Park, Jake Hyun, Hojoon Kim, and Jae W. Lee. DecDEC: A Systems Approach to Advancing Low-Bit LLM Quantization. 19th USENIX Symposium on Operating Systems Design and Implementation (OSDI 25), 2025. https://www.usenix.org/system/files/osdi25-park-yeonhong.pdf.

Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache. ICML 2024. https://arxiv.org/abs/2402.02750.

Tengxuan Liu, Shiyao Li, Jiayi Yang, Tianchen Zhao, Feng Zhou, Xiaohui Song, Guohao Dai, Shengen Yan, Huazhong Yang, and Yu Wang. PM-KVQ: Progressive Mixed-precision KV Cache Quantization for Long-CoT LLMs. arXiv:2505.18610, 2025. https://arxiv.org/abs/2505.18610.

Qingyue Yang, Jie Wang, Xing Li, Zhihai Wang, Chen Chen, Lei Chen, Xianzhi Yu, Wulong Liu, Jianye Hao, Mingxuan Yuan, and Bin Li. AttentionPredictor: Temporal Patterns Matter for KV Cache Compression. NeurIPS 2025. https://arxiv.org/abs/2502.04077.

Hao Mark Chen, Fuwen Tan, Alexandros Kouris, Royson Lee, Hongxiang Fan, and Stylianos I. Venieris. Progressive Mixed-Precision Decoding for Efficient LLM Inference. arXiv:2410.13461, revised February 2025. https://arxiv.org/abs/2410.13461.

Yinglong Li, Xiaoyu Liu, Jiacheng Li, Ruikang Xu, Yinda Chen, and Zhiwei Xiong. QMamba: Post-Training Quantization for Vision State Space Models. arXiv:2501.13624, January 2025. https://arxiv.org/abs/2501.13624.

Akshat Ramachandran, Mingyu Lee, Huan Xu, Souvik Kundu, and Tushar Krishna. OuroMamba: A Data-Free Quantization Framework for Vision Mamba. arXiv:2503.10959, ICCV 2025. https://arxiv.org/abs/2503.10959.

Alessandro Pierro and Steven Abreu. Mamba-PTQ: Outlier Channels in Recurrent Large Language Models. arXiv:2407.12397, 2024. https://arxiv.org/abs/2407.12397.

Hung-Yueh Chiang, Chi-Chih Chang, Natalia Frumkin, Kai-Chiang Wu, and Diana Marculescu. Quamba: A Post-Training Quantization Recipe for Selective State Space Models. arXiv:2410.13229, 2024. https://arxiv.org/abs/2410.13229.

Hung-Yueh Chiang, Chi-Chih Chang, Natalia Frumkin, Kai-Chiang Wu, Mohamed S. Abdelfattah, and Diana Marculescu. Quamba2: A Robust and Scalable Post-training Quantization Framework for Selective State Space Models. arXiv:2503.22879, 2025. https://arxiv.org/abs/2503.22879.

Zukang Xu, Yuxuan Yue, Xing Hu, Zhihang Yuan, Zixu Jiang, Zhixuan Chen, Jiangyong Yu, Chen Xu, Sifan Zhou, and Dawei Yang. MambaQuant: Quantizing the Mamba Family with Variance Aligned Rotation Methods. arXiv:2501.13484, 2025. https://arxiv.org/abs/2501.13484.

Yizhi Chen and Ahmed Hemani. Late Breaking Results: Quamba-SE: Soft-edge Quantizer for Activations in State Space Models. arXiv:2601.09451, DATE Late Breaking Results 2026. https://arxiv.org/abs/2601.09451.

Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. arXiv:2211.10438, ICML 2023. https://arxiv.org/abs/2211.10438.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration. arXiv:2306.00978, MLSys 2024. https://arxiv.org/abs/2306.00978.

Saleh Ashkboos, Amirkeivan Mohtashami, Maximilian L. Croci, Bo Li, Pashmina Cameron, Martin Jaggi, Dan Alistarh, Torsten Hoefler, and James Hensman. QuaRot: Outlier-Free 4-Bit Inference in Rotated LLMs. arXiv:2404.00456, 2024. https://arxiv.org/abs/2404.00456.

Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. KVQuant: Towards 10 Million Context Length LLM Inference with KV Cache Quantization. arXiv:2401.18079, NeurIPS 2024. https://arxiv.org/abs/2401.18079.

Wonsuk Jang and Thierry Tambe. BlockDialect: Block-wise Fine-grained Mixed Format Quantization for Energy-Efficient LLM Inference. arXiv:2501.01144, ICML 2025. https://arxiv.org/abs/2501.01144.

Yesheng Liang, Haisheng Chen, Song Han, and Zhijian Liu. ParoQuant: Pairwise Rotation Quantization for Efficient Reasoning LLM Inference. arXiv:2511.10645, 2025. https://arxiv.org/abs/2511.10645.

Qwen Team. Qwen3.6-35B-A3B model card. Hugging Face, 2026. https://huggingface.co/Qwen/Qwen3.6-35B-A3B.

Songlin Yang, Bailin Wang, Yikang Shen, Rameswar Panda, and Yoon Kim. Gated Linear Attention Transformers with Hardware-Efficient Training. arXiv:2312.06635, 2023. https://arxiv.org/abs/2312.06635.

Songlin Yang, Jan Kautz, and Ali Hatamizadeh. Gated Delta Networks: Improving Mamba2 with Delta Rule. arXiv:2412.06464, 2024. https://arxiv.org/abs/2412.06464.

Moonshot AI Kimi Team: Yu Zhang et al. Kimi Linear: An Expressive, Efficient Attention Architecture. arXiv:2510.26692, October 2025. https://arxiv.org/abs/2510.26692.